

FNU Avina

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Education

New York University, *MS in Computer Science* | GPA: 3.6 | New York, NY May 2025
Vellore Institute of Technology, *B.Tech in Computer Science and Engineering* | GPA: 3.7 | India June 2021

Skills

Programming Languages: Java, Python, Kotlin, C++ , C , Javascript, Typescript
Backend & APIs: : Spring Boot, RESTful APIs, Microservices, Django, Flask, Node.js, JDBC, Spring JPA, JWT, junit
Database & BigData: SQL, MySQL, MongoDB, PostgreSQL, DynamoDB, Supabase, SQLite, Kafka, Hadoop, Spark
Frontend & Mobile : HTML, CSS, PHP, React Native
Cloud & DevOps: AWS (S3, Lambda, API gateway, SQS, EC2), GCP, Docker, Kubernetes, Git, Github Actions
AI & Machine Learning: scikit-learn, Keras, PyTorch, TensorFlow, NLP, Deep Learning, PyCaret, RAG, LLM

Experience

AI Systems Engineer, QuantAI – New York, NY May 2025 – Present

- Leading development of ALX, novel compiler for domain-specific language used in AI-driven high-frequency trading system
- Engineering the core Syntax-to-Vectorization Pathway in MLIR and final code emission via the LLVM backend into highly-optimized SIMD machine code, that outperformed Pandas and NumPy by 2.5X in benchmarks
- Received NYU patent for vectorization and loop-free concepts in the compiler for high-frequency systems

Research Intern, QuantAI – New York, NY Feb 2025 – Present

- Achieved 100x speed improvement in complex trading algorithms by implementing low-level CPU optimizations, memory management, and assembly-level programming techniques in C, Assembly and C++
- Developing an AI trading co-pilot where specialized LLM agents collaborate to research markets, devise algorithmic strategies, and simulate performance

Engineering Manager Intern, Develop for Good – New York, NY Oct 2024 – May 2025

- Led a team of 9 engineers developing a mobile app using React Native, TypeScript, Supabase and Expo
- Implemented core features including home page, navigation, tagging, filtering and search functionality

Backend Engineer, Pharameasy – Bangalore, India July 2021 – July 2023

- Designed and built scalable Platform Order Management System, integrating warehouse systems through RESTful APIs, unifying multiple business verticals while maintaining 100% uptime for millions of users
- Streamlined Kubernetes deployments for microservices architecture built on Spring Boot, Java and Kotlin
- Reduced warehouse turnaround time by 37% and operational costs by 20% through Pick Path Optimization and Smart Binning algorithms, utilizing real-time data streams from AWS via Kafka
- Rebuilt returns management system end-to-end, increasing returnable items by 35% and enabling resale of returned inventory, directly contributing to revenue recovery and loss reduction across cross-functional teams
- Resolved critical production incidents affecting millions of users, reducing system downtime by 30% and decreasing e-way bill failures from 35% to 3%, resulting in \$2.1M annual cost savings

Research Intern (Image Processing), ISRO Space Applications Center – Bangalore, India Dec 2020 – June 2021

- Developed a new algorithm for incidence angle normalization of Geospatial and Radar satellite images using Python, reducing amplitude by 5X and backscatter by 10dB, enhancing remote sensing accuracy

Projects

Classmate.io Dec 2023

- Gamified learning platform with automated quiz generation, lecture transcription, and text summarization using 13+ AWS services (Lambda, SageMaker, S3, DynamoDB, API Gateway, Transcribe)
- Implemented MLOps pipeline with SageMaker for automated model training and deployment; fine-tuning DistilBart-CNN on Cornell arXiv dataset for text summarization
- Developed MCQ generation using NLP (NER, Word2Vec, GloVe embeddings) with smart distractor generation

VectorMind July 2025

- RAG-powered chatbot with Gemini 2.0 Flash LLM, FAISS vector search, HuggingFace embedding and custom LangChain implementation achieving <500ms query latency